HH-I-574B
October 16, 1974
SUPERSEDING
Fed. Spec. HH-I-574A
May 31, 1966

FEDERAL SPECIFICATION

INSULATION, THERMAL (PERLITE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an expanded perlite loose fill thermal insulating material in dry condition for use as fill insulation to restrict the flow of heat in the temperature range from - 50 to 1400°F (-45 to 760°C).

1.2 Classification. Perlite loose fill insulation covered by this specification shall be of one type.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge Polyolefin.
O-T-620 - 1,1,1 - Trichloroethane, Technical Inhibited (Methyl Chloroform).
UU-S-48 - Sacks, Shipping, Paper.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unlwss a specific issue is identified, the issue in effect on date of invitation for bids or request for proporal shall apply.

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American Society for Testing and Materials (ASTM) Standards:

C 136 - Method of Test for Seive or Screen Analysis of Fine and Course Aggregates.
C 177 - Method of Test for Thermal Conductivity of Materials by Means of the Guarded Hot Plate.
C 520 - Method of Test for Density of Granular Loose Fill Insulation.
C 549 - Specification for Perlite Loose Fill Insulation.

(Application for copies should be addressed the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Expanded perlite loose fill insulation shall be produced by the expanding of natural perlite ore by heating and shall be treated to produce a specific property such as water repellency, and shall meet the requirements of ASTM C 549.

3.2 Water repellency. When tested in accordance with 4.4.3, the water repellency of perlite shall not be less than 175 milliliters (Ml.) of water repelled.

3.3 Combustibility. When tested in accordance with 4.4.5, perlite shall not spark, burn, or smoke.

3.4 Solvent solubles. When tested in accordance with 4.4.6, the perlite shall not contain more than 0.1 percent by weight of solvent solubles.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all perlite submitted for inspection at the same time.

4.2.2 Sampling for tests. Samples for tests shall be in accordance with level S-2 of MIL-STD-105. The acceptable quality level (AQL) shall be 2.5 percent defective for specimens used in each test.

4.3 Examination.

4.3.1 End item. Perlite shall be examined for defects listed in table I. The acceptable quality levels (AQL's) in MIL-STD-105 shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE I. Classification of defects, end item

Defects	Major	Minor
Bags not fully closed.	X	
Bags split or torn.	X	
Holes in bags.	X	
Dirty or damp insulation.	X	
Marking omitted, incomplete, illegible, or in wrong location.		X

4.4 Test methods.

4.4.1 Density. Density shall be determined in accordance with ASTM C 520, (see 3.1).

4.4.2 Thermal conductivity. Thermal conductivity shall be determined in accordance with ASTM C 177, (see 3.1).

4.4.3 Water repellency. The water repellency of perlite insulation shall be determined in accordance with the following procedure, (see 3.2). A rigid plastic tube 2 inch ID by 12 inches long with a 16 mesh plastic screen covering firmly fastened or adhered to the bottom opening shall be employed. The tube shall be marked at 7.77 inches from screen covered end indicating a volume of 400 ml. The plastic tube shall be filled with the perlite insulation to level slightly above the 400-ml. mark. It shall then be compacted by dropping the tube from a height of approximately 3 inches on a large rubber stopper for 10 drops. As the sample compacts to a level below the 400-ml. mark, additional material shall be added so that after the tenth drop the level of the sample is within 1/8 inch of the 400-ml. mark. The tube shall be kept in a vertical position during compaction. With the tube supported in a vertical position and a beaker positioned under the tube, rapidly pour 250 ml. of water onto the perlite insulation. Allow 3 minutes for the water to drain through specimen, then tilt tube at approximately 45° to drain water collected on screen. Pour water collected in beaker into graduate and record amount repelled by the treated perlite loose fill insulation. At least 3 tests shall be conducted using separate samples of the perlite insulation and the results averaged.

4.4.4 Grading. Particle size of perlite shall be determined in accordance with ASTM C 136, except that when a mechanical sieving device is used, the sieving time shall be 5 minutes and the test sample shall be 50 grams (g.) of material, (see 3.1).

4.4.5 Combustibility. Perlite loose fill insulation shall be held in a suitable fixture and positioned in direct contact with the flame of a Bunsen burner. For 15 seconds (see 3.3).

4.4.6 Solvent solubles. Place 10 g. of perlite in a beaker, then add 100 cubic centimeters of trichloroethane in conformance with O-T-620. Stir for 3 minutes then filter through a Whatman 42 paper into a beaker previously weighed to 0.001 g. Evaporate the trichloroethane on a steam bath. Heat in an oven for 30 minutes at 105° $\pm$ 2°C. Cool in a desiccator and calculate the decrease in weight (see 3.4).

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Loose fill perlite insulation shall be packed in quanitites as specified in the contract or purchase order in sacks conforming to UU-S-48, level A, 6-6x, or in plastic bags constructed from material conforming to L-P-378, a minimum of 4 mils thick. The plastic bags shall be heat sealed after filling. Excess air within the filled bag shall be kept to a practicable minimum. When the UU-S-48 sack is used, the gross weight shall not exceed 40 pounds.

5.1.2 Level B. Loose fill insulation shall be packed in quanitites as specified in the contract or purchase order in sacks conforming to UU-S-48, level B, 3-3x, or in plastic bags constructed from material conforming to L-P-378, a minimum of 4 mils thick. The plastic bags shall be heat sealed after filling. Excess air within the filled bag shall be kept to a practicable minimum. When the UU-S-48 sack is used, the gross weight shall not exceed 40 pounds.

5.1.3 Level C. Perlite insulation shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification rules and the National Motor Freight Classification rules.

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5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the bags and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to marking required by the contract or order, the bags and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Thermal conductivity "K" factor 0.38 at a mean temperature of 75°F. is applicable to building construction. Thermal conductivity "K" factors 0.49 and above are intended for high temperature applications.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of lot, if different (see 4.2.1).
(c) Level of packing required (see 5.1).
(d) Whether test are required.

6.3 Description of perlite. Perlite is an inert siliceous volcanic rock containing a small amount of combined water. When crude perlite particles are heated above 1600°F., the combined water vaporizes and the perlite expands and turns white. Expanded perlite contains thousands of macroscopic air cells or voids which account for its thermal insulation and light weight.

6.4 Coverage. The approximate coverage of perlite insulation for masonry walls can be determined in accordance with table II.

TABLE II. Approximate coverage for masonry walls

Wall area sq. ft.	Number of bags (4 cu. ft.) required: Core fill - block size			Cavity fill - cavity width			
	6 inch	8 inch	12 inch	1 inch	2 inch	2-1/2 inch	3 inch
100	5	7	12	2	4	5	6
500	23	33	58	10	21	26	31
1000	46	65	118	21	42	52	62
2000	91	130	236	42	84	104	125
3000	137	195	354	62	124	155	187

MILITARY CUSTODIANS:

Navy - YD
Air Force - 84

User activities:

Navy - MC
Army - CE

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

COMMERCE - NBS
GSA - FSS, PCD
HUD - TCS
INTERIOR - BPA

Military Coordinating Activity:

Navy - YD

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 25 cents each.

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